



ROUTER CONFIGURATION AND ROUTING PROTOCOLS USING CISCO PACKET TRACER

Practical Assignment Report

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Module: Computer Networks

Task 1 – Router Basic Configuration

1. Configuration Commands Used

```
enable  
  
configure terminal  
  
hostname KandyNSBM_39148  
  
line console 0  
  
password NSBM  
  
login  
  
end
```

2. Screenshot – Configured Hostname

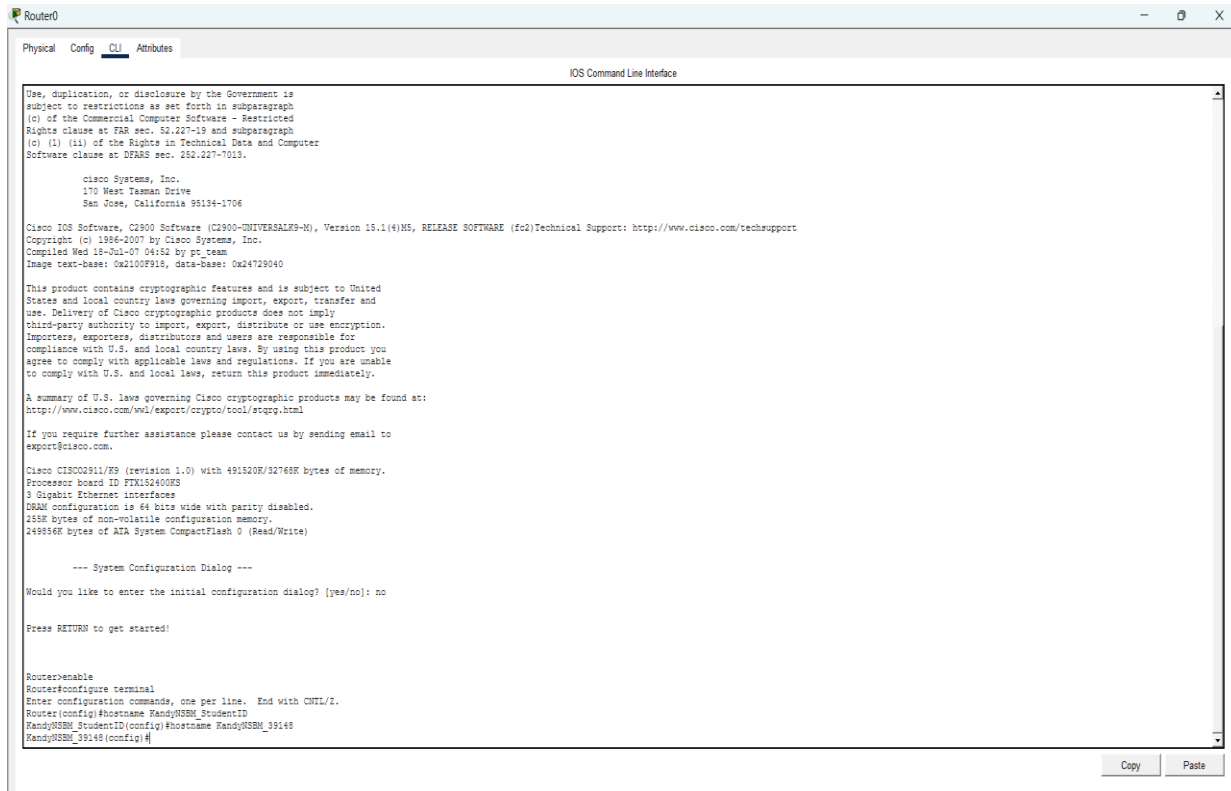


Figure 1: Configured Router Hostname

3. Screenshot – Console Password Configuration

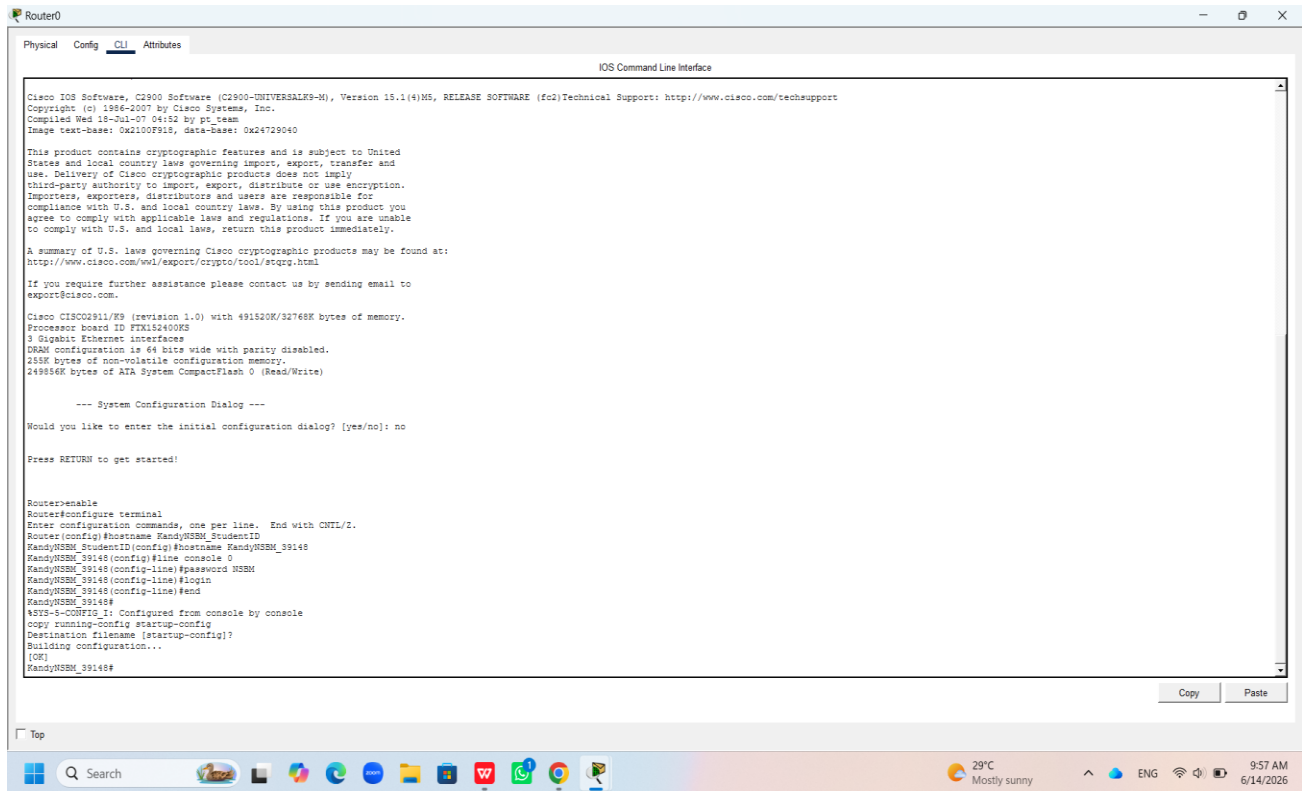


Figure 2: Console Password Configuration

4. Brief Explanation

The hostname command was used to assign a unique name to the router. The line console 0 command was used to access console line configuration mode. The password command assigned the console password as NSBM, while the login command enabled password verification when accessing the router through the console port.

Task 2 – Static Routing Configuration

1. Network Topology

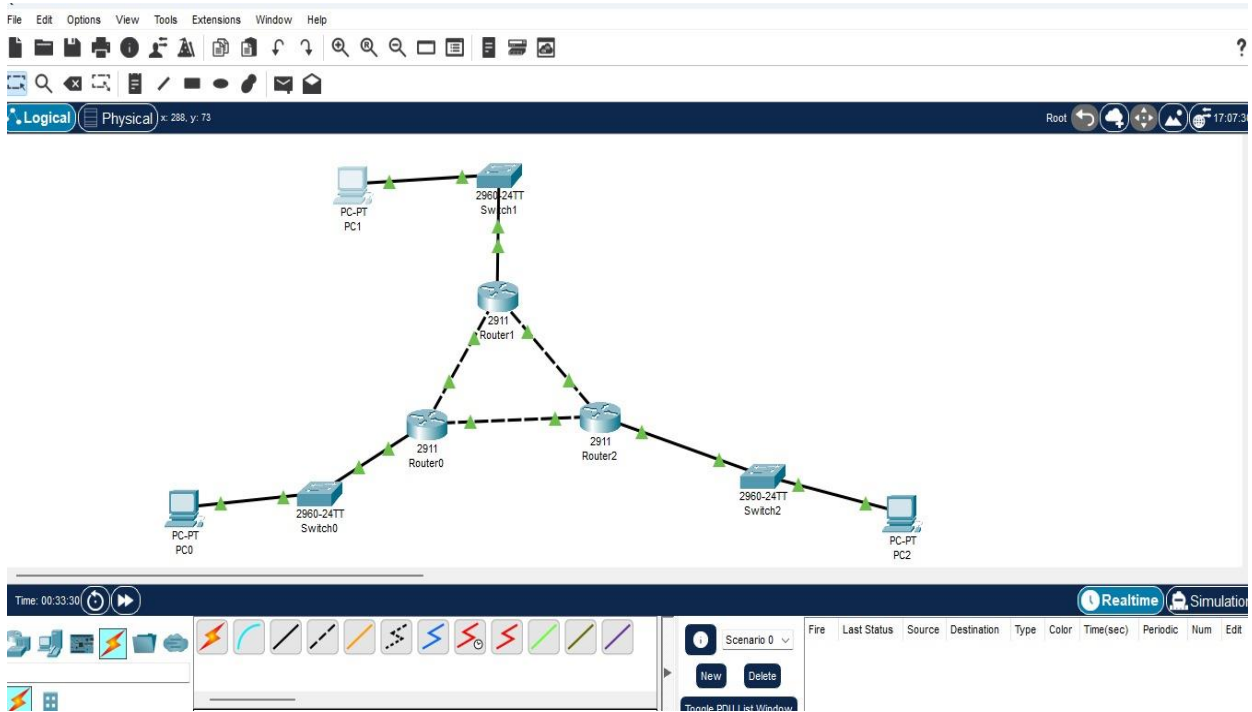


Figure 1: Static Routing Network Topology

IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Purpose
Router0	G0/0	192.168.1.1	255.255.255.0	LAN
Router0	G0/1	10.0.0.1	255.255.255.252	Router Link
Router1	G0/0	10.0.0.2	255.255.255.252	Router Link
Router1	G0/1	11.0.0.1	255.255.255.252	Router Link

Router2	G0/0	11.0.0.2	255.255.255.252	Router Link
Router2	G0/1	192.168.2.1	255.255.255.0	LAN

2. Static Route Configuration Commands

- Router0

```
ip route 11.0.0.0 255.255.255.252 10.0.0.2
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

- Router1

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
ip route 192.168.2.0 255.255.255.0 11.0.0.2
```

- Router2

```
ip route 10.0.0.0 255.255.255.252 11.0.0.1
ip route 192.168.1.0 255.255.255.0 11.0.0.1
```

3. Routing Table Screenshots

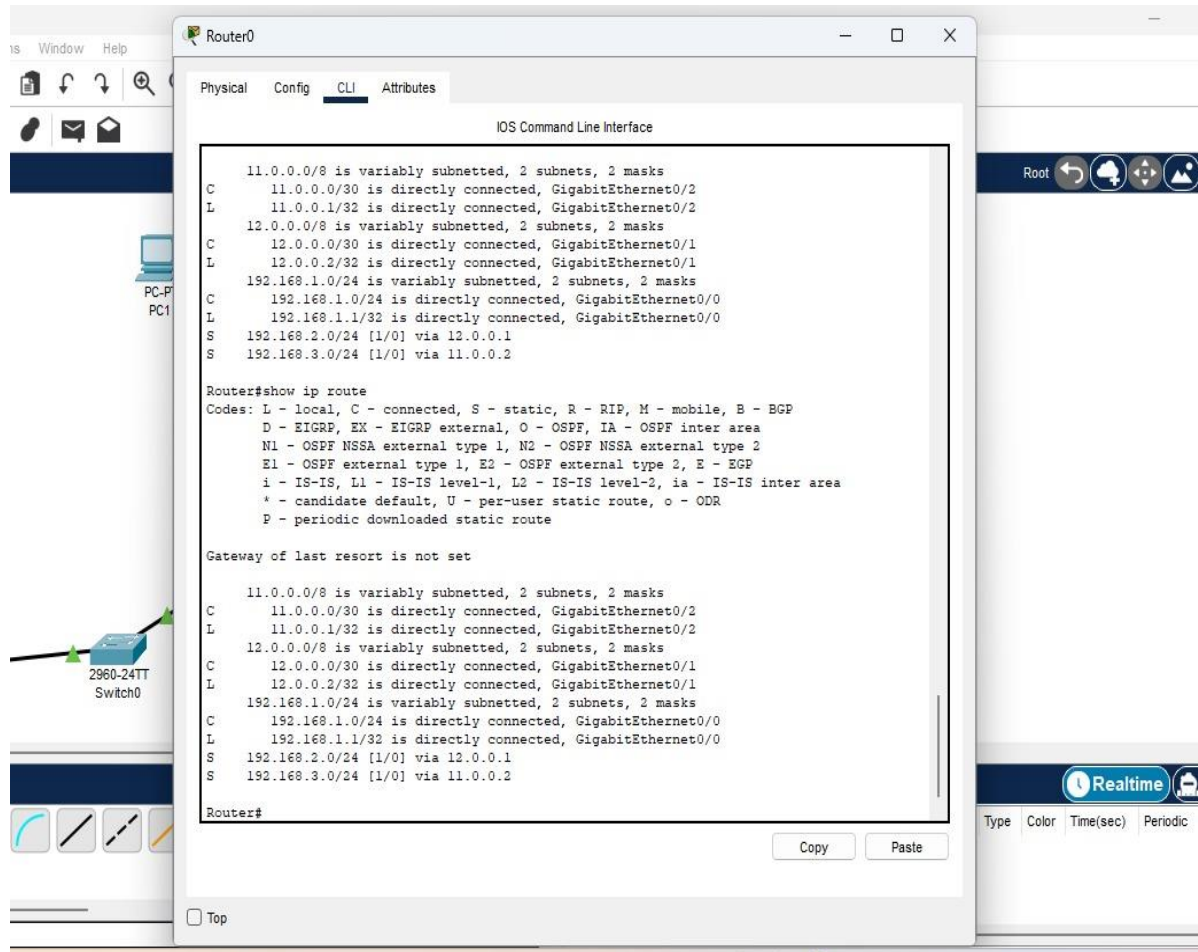


Figure 2: Router0 Routing Table

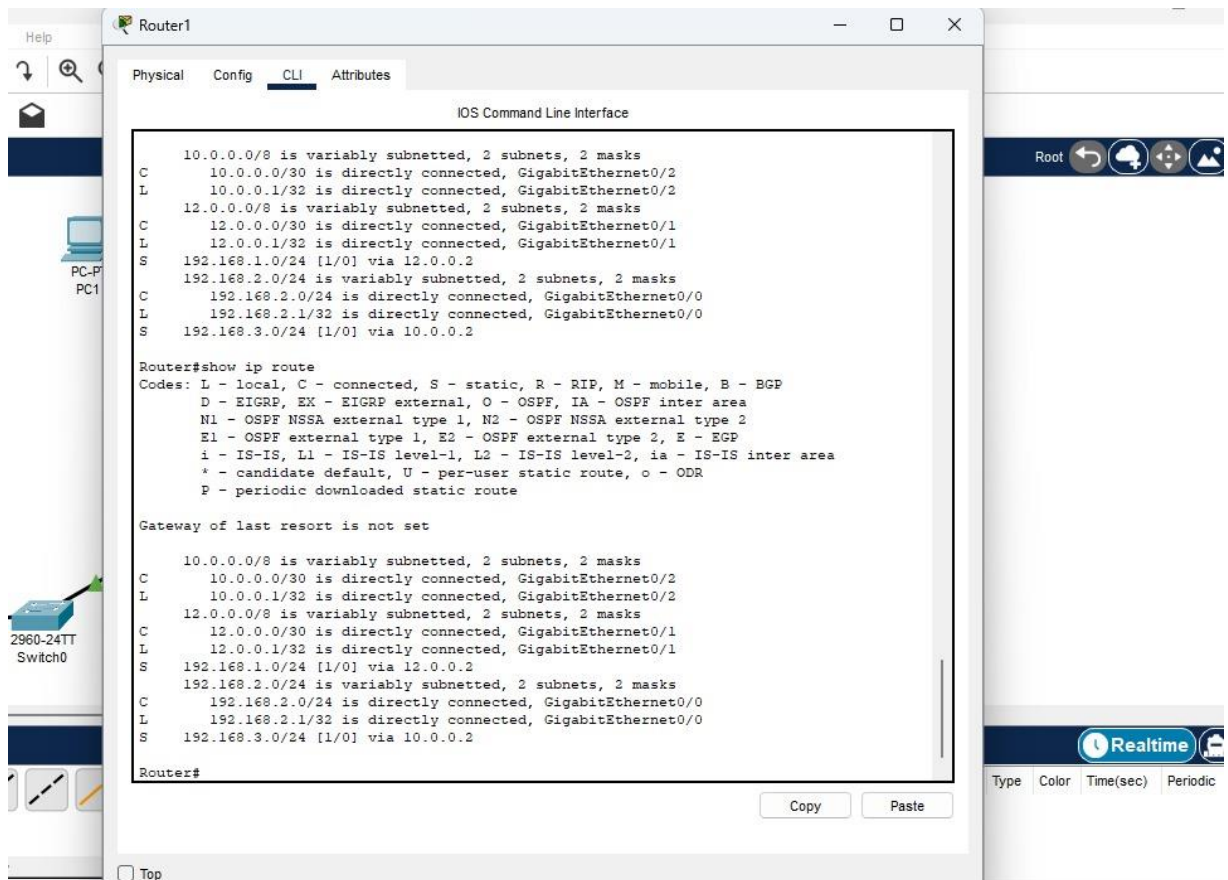


Figure 3: Router1 Routing Table

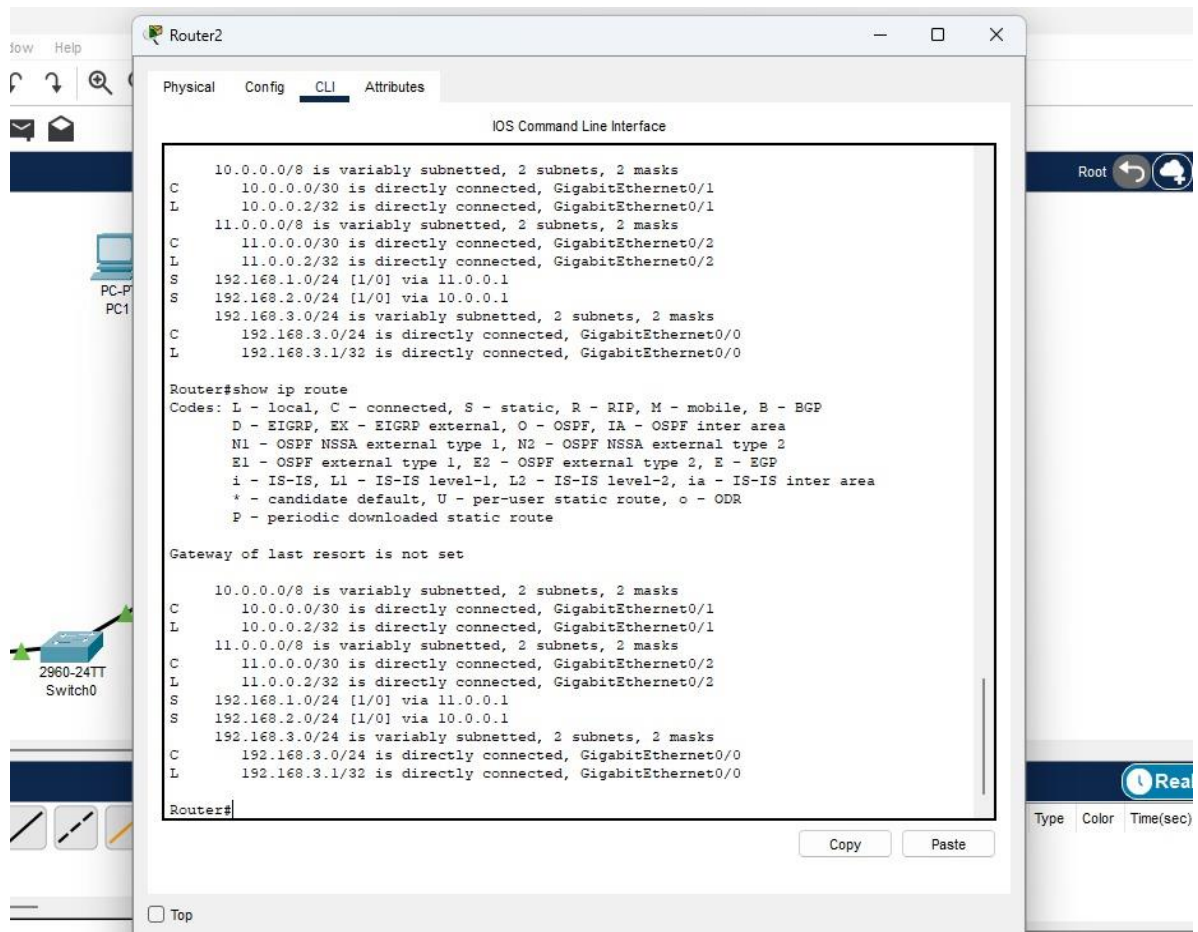


Figure 4: Router2 Routing Table

4. Ping Test Result

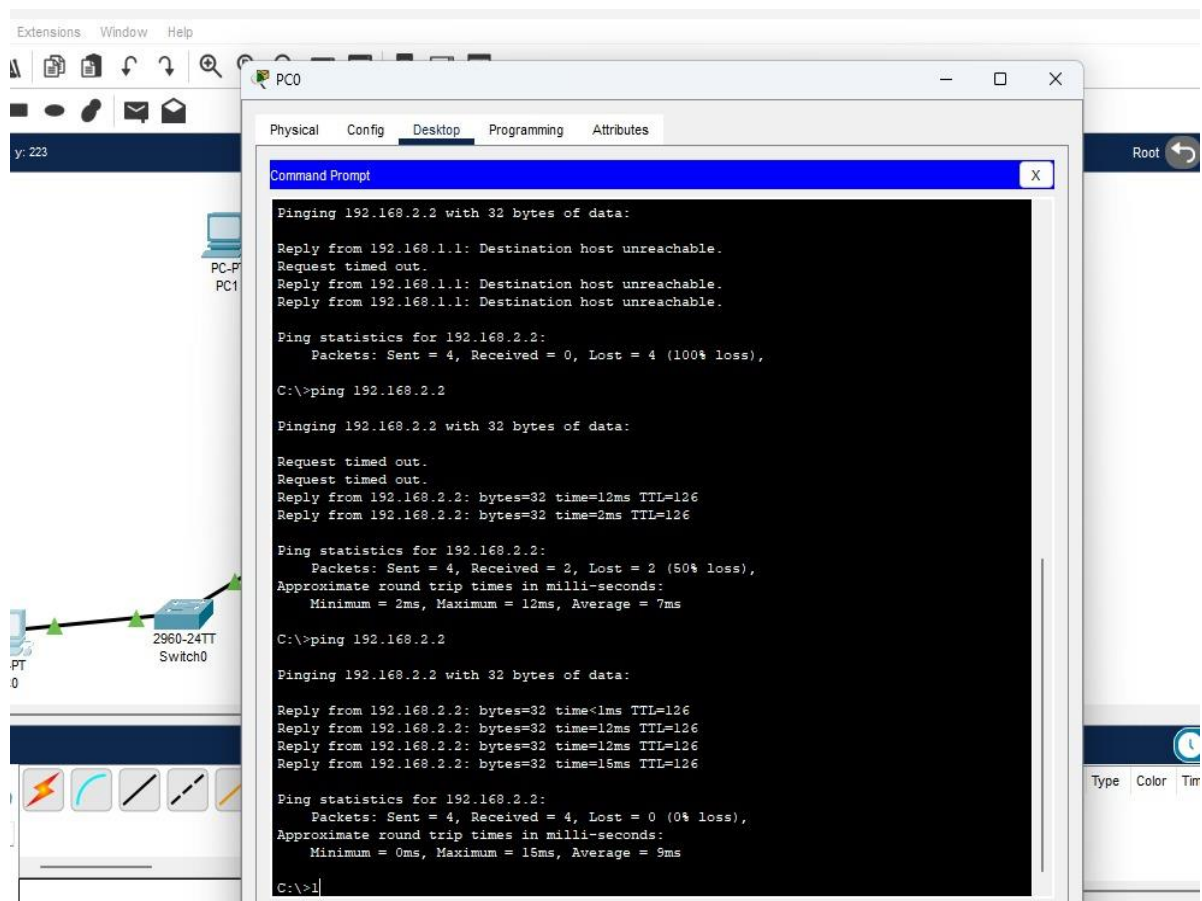


Figure 5: Successful Ping from PC0 to PC1

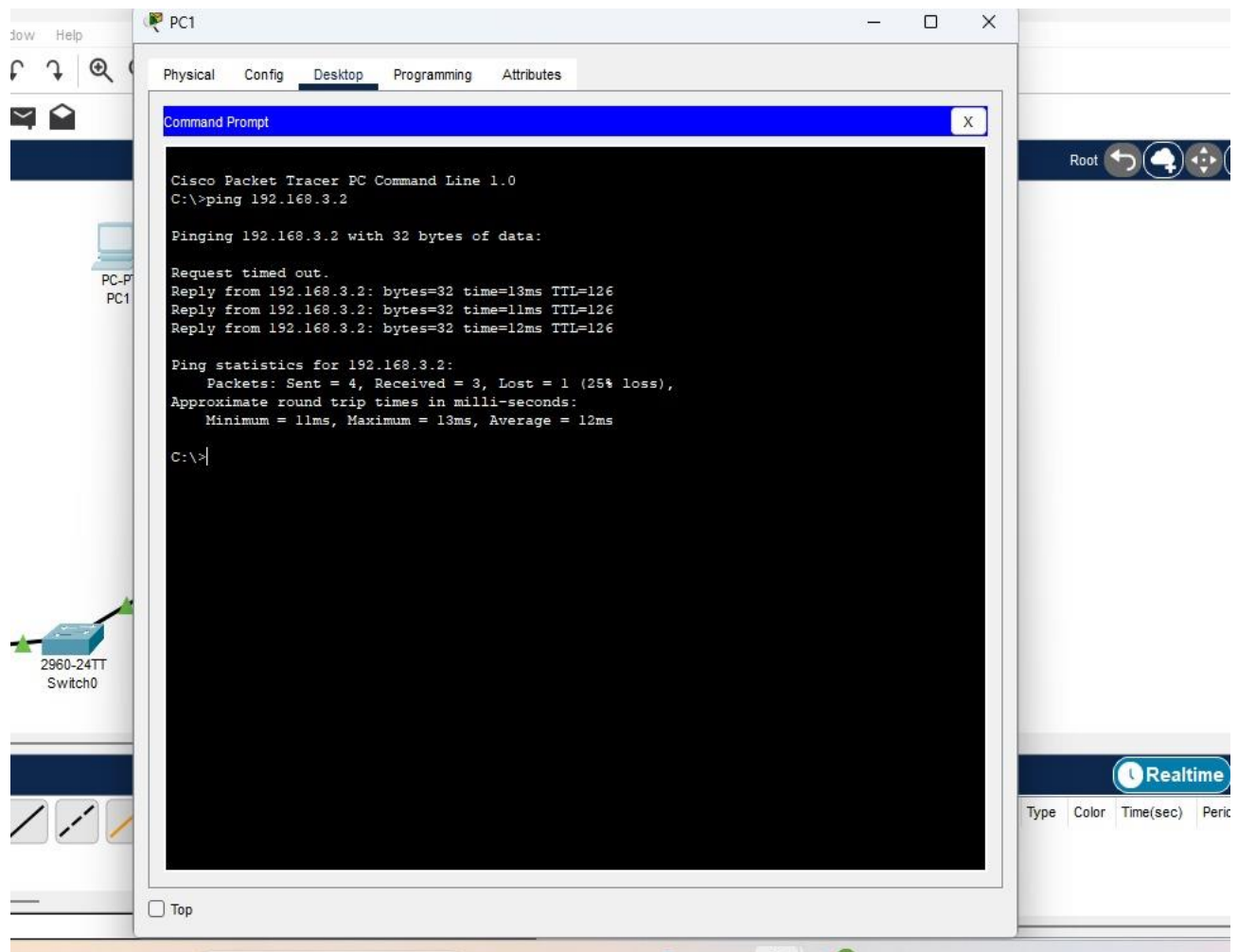


Figure 6: Successful Ping from PC1 to PC0

5. Brief Explanation

Static routing was configured on all routers to enable communication between different networks. The 'ip route' command was used to manually define routes to remote networks. Interface IP addresses were assigned using the 'ip address' command, while 'no shutdown' was used to activate router interfaces. The 'show ip route' command was used to verify the routing tables and confirm that static routes were successfully installed. Finally, connectivity between networks was tested using the 'ping' command, which confirmed successful communication between all devices.

Task 3 – Dynamic Routing Configuration

1. Network Diagram

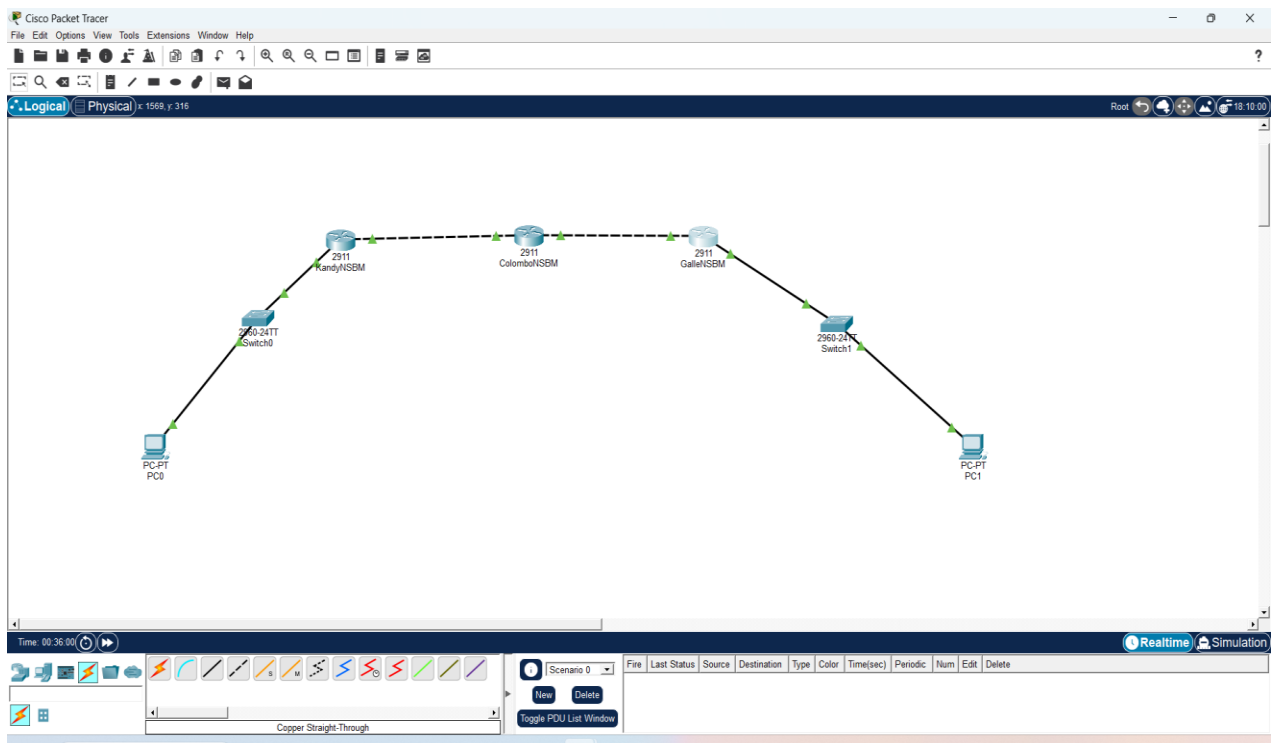


Figure 7: Dynamic Routing Network Topology

Part A – RIP Routing

2. RIP Configuration Commands

- KandyNSBM

```
router rip
version 2
network 172.16.0.0
network 14.0.0.0
no auto-summary
```

- ColomboNSBM

```
router rip
version 2
network 14.0.0.0
network 15.0.0.0
no auto-summary
```

- GalleNSBM

```
router rip
version 2
network 172.16.0.0
network 15.0.0.0
no auto-summary
```

3. Screenshot of Routing protocol configuration

```
KandyNSBM
Physical Config CLI Attributes
IOS Command Line Interface

http://www.cisco.com/wvl/export/crypto/tool/stggy.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/F9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400S
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINE-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINE-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
end
Router#
A9Y5-5-CONFIG_1: Configured from console by console
show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up
GigabitEthernet0/1 10.0.0.1 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#end
Translating "end"...domain server (255.255.255.255)

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Top
```

Figure 8: KandyNSBM Routing protocol configuration

```
ColomboNSBM
Physical Config CLI Attributes
IOS Command Line Interface

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/F9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400S
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Press RETURN to get started!

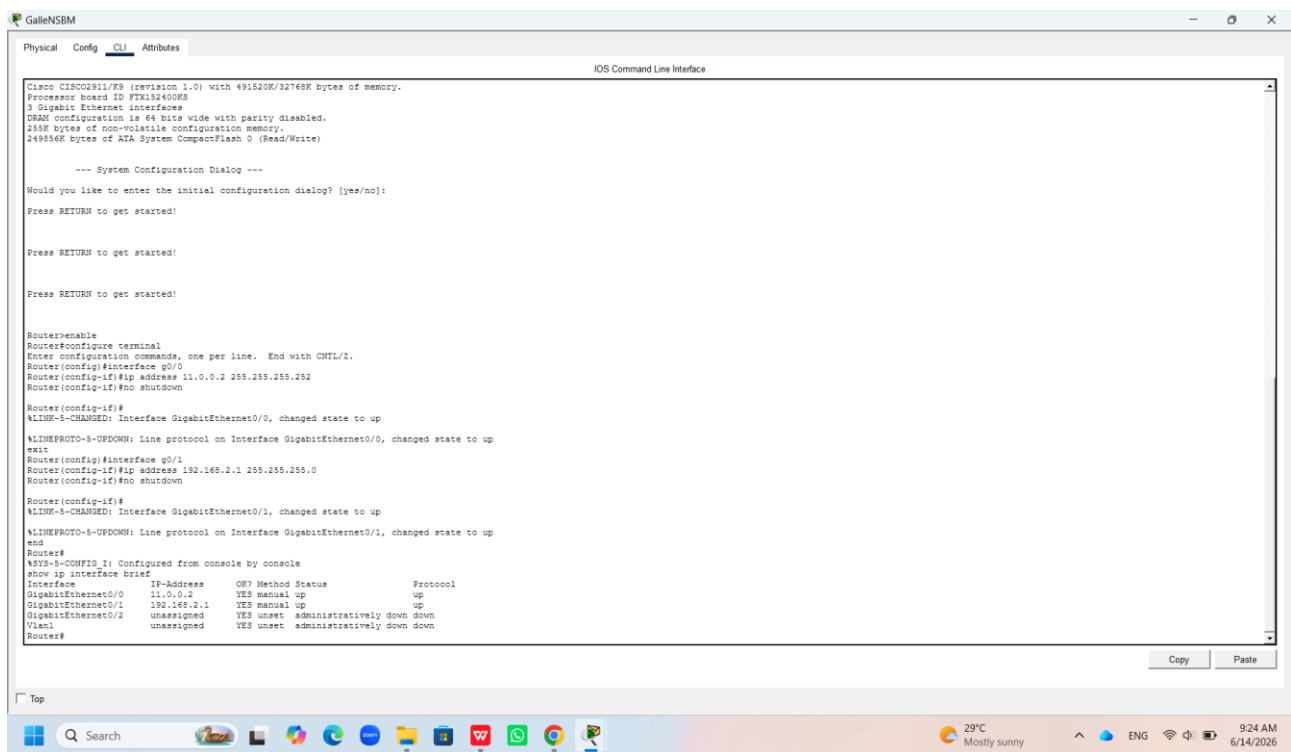
Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINE-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINE-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
end
Router#
A9Y5-5-CONFIG_1: Configured from console by console
show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.0.2 YES manual up
GigabitEthernet0/1 11.0.0.1 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#

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```

Figure 9: ColomboNSBM Routing protocol configuration



The screenshot displays the GalleNSBM IOS Command Line Interface. The window title is "GalleNSBM" and it has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
Cisco CISC02911/99 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400K
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Press RETURN to get started!

Press RETURN to get started!

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 11.0.0.2 255.255.255.252
Router(config-if)#no shutdown

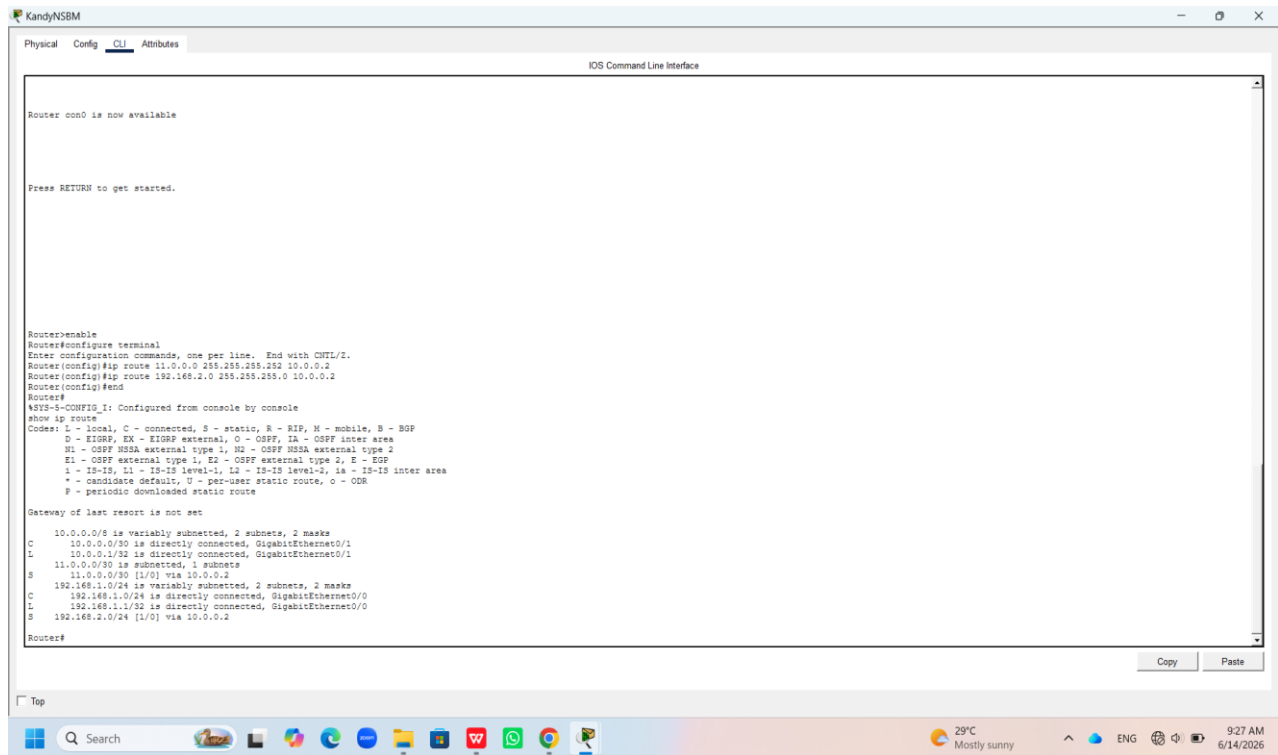
Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 11.0.0.2 YES manual up up
GigabitEthernet0/1 192.168.2.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

The bottom of the window shows a Windows taskbar with various icons, including the Start button, Search, and several application icons. The system tray on the right shows the temperature (29°C), weather (Mostly sunny), language (ENG), and the date/time (9:24 AM, 6/14/2026).

Figure 10: GalleNSBM Routing protocol configuration

4. Screenshots of routing tables



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 11.0.0.0 255.255.255.252 10.0.0.2
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I- Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
S       11.0.0.0/30 is subnetted, 1 subnets
        11.0.0.0/30 [1/0] via 10.0.0.2
S       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S       192.168.2.0/24 [1/0] via 10.0.0.2

Router#
```

Figure 11: KandyNSBM Routing table

ColomboNSBM

Physical Config CLI Attributes

IOS Command Line Interface

```
!LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
!LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
end
Router#
#SIS-5-CONFIG_1: Configured from console by console
show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.0.2        YES manual up          up
GigabitEthernet0/1  11.0.0.1        YES manual up          down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.2.0 255.255.255.0 11.0.0.2
Router(config)#end
Router#
#SIS-5-CONFIG_1: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.2/32 is directly connected, GigabitEthernet0/0
C    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    11.0.0.0/30 is directly connected, GigabitEthernet0/1
L    11.0.0.1/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.0.0.1
S    192.168.2.0/24 [1/0] via 11.0.0.2
Router#
```

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Windows taskbar: Search, 29°C Mostly sunny, 9:28 AM 6/14/2026

Figure 12: ColomboNSBM Routing table

GalleNSBM

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown
Router(config-if)#
!LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
!LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
end
Router#
#SIS-5-CONFIG_1: Configured from console by console
show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  11.0.0.2        YES manual up          up
GigabitEthernet0/1  192.168.2.1     YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.255.255.252 11.0.0.1
Router(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
Router(config)#end
Router#
#SIS-5-CONFIG_1: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
S    10.0.0.0/30 [1/0] via 11.0.0.1
C    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    11.0.0.0/30 is directly connected, GigabitEthernet0/0
L    11.0.0.2/32 is directly connected, GigabitEthernet0/0
S    192.168.1.0/24 [1/0] via 11.0.0.1
S    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/1
L    192.168.2.1/32 is directly connected, GigabitEthernet0/1
Router#
```

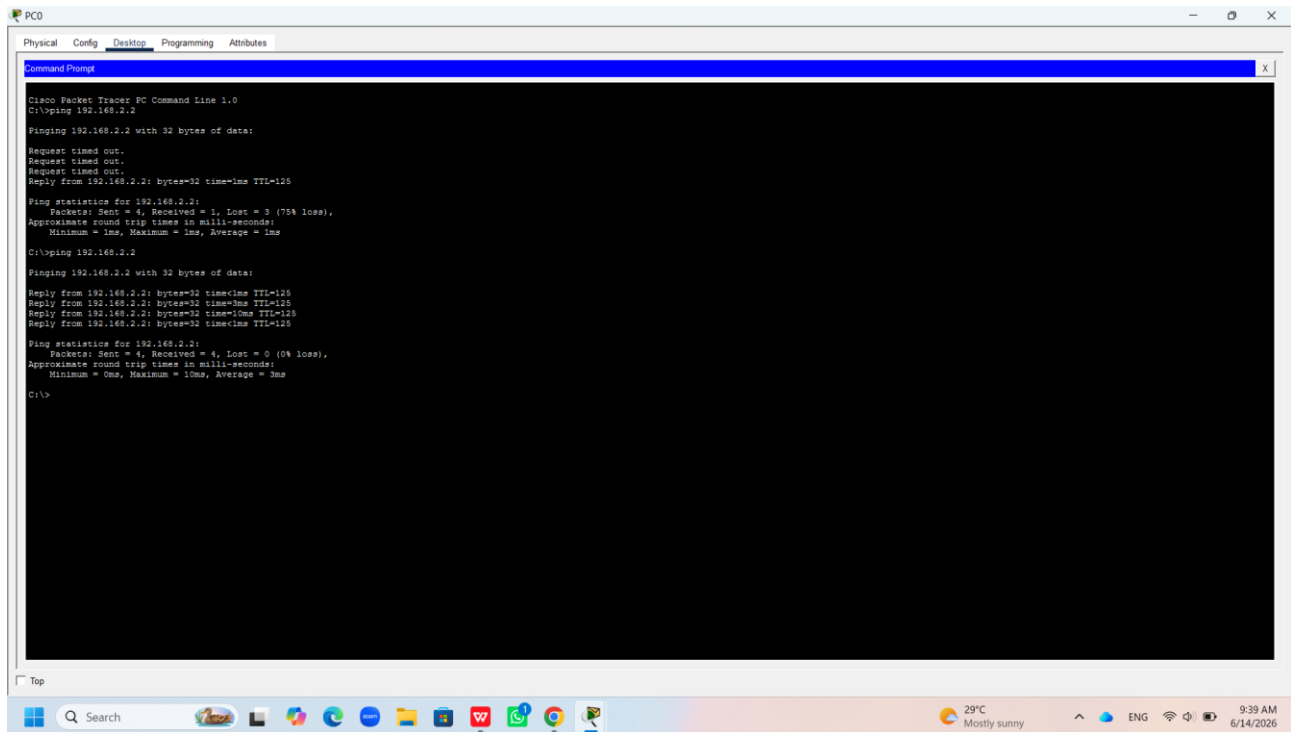
Copy Paste

Top

Windows taskbar: Search, 29°C Mostly sunny, 9:29 AM 6/14/2026

Figure 13: GalleNSBM Routing table

5. PIN Test Result



The screenshot shows a Windows desktop environment with a 'PC0' window open. Inside the window is a 'Command Prompt' titled 'Cisco Packet Tracer PC Command Line 1.0'. The command prompt shows the execution of a ping command to 192.168.2.2. The first attempt shows three 'Request timed out.' messages followed by a successful reply from 192.168.2.2 with 32 bytes of data, time=1ms, and TTL=125. The statistics for this attempt show 4 packets sent, 1 received, and 3 lost (75% loss). The second attempt shows four successful replies from 192.168.2.2 with 32 bytes of data, times of 3ms, 3ms, 10ms, and 125ms, all with TTL=125. The statistics for this attempt show 4 packets sent, 4 received, and 0 lost (0% loss). The window has a taskbar at the bottom with various application icons and a system tray showing the date and time as 9:39 AM on 6/14/2026.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=10ms TTL=125
Reply from 192.168.2.2: bytes=32 time=125ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Figure 14: PC0 PIN Test result

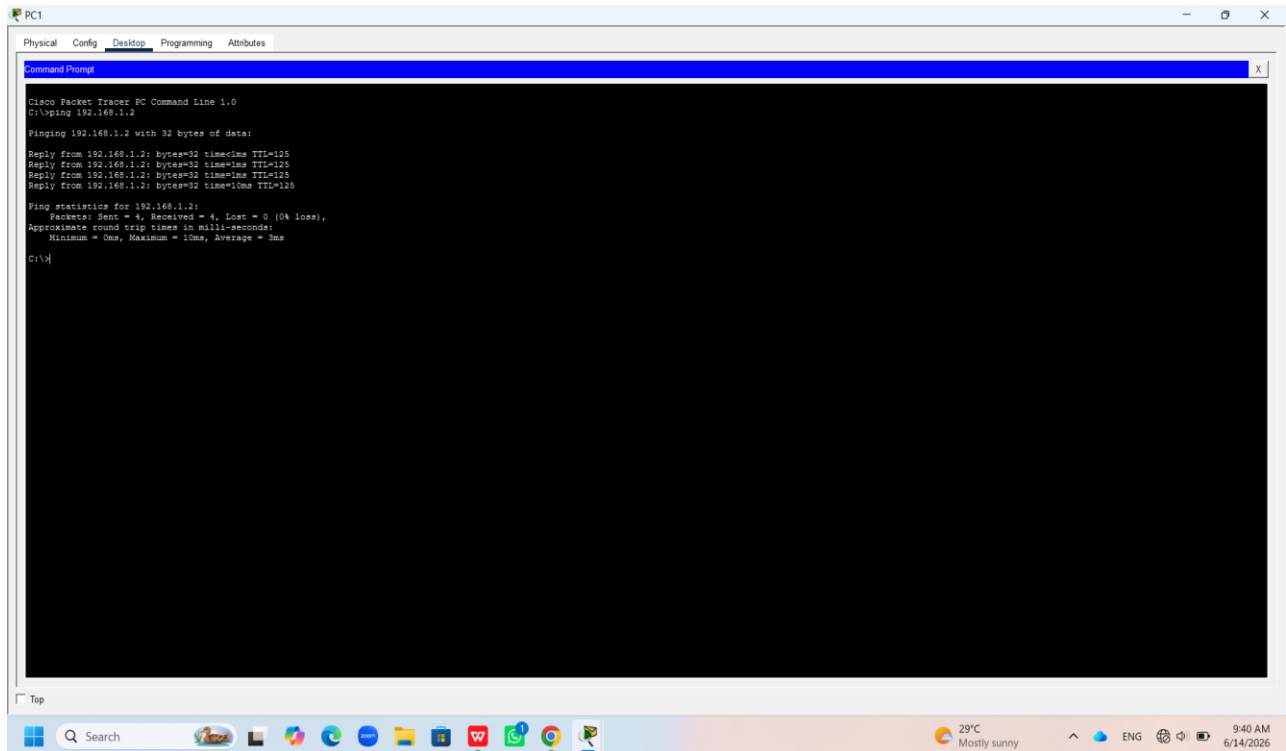


Figure 15: PC1 PIN Test result

6. Brief Explanation of Commands Used

Dynamic routing was implemented using both RIP version 2 and EIGRP routing protocols. RIP was configured by enabling RIP version 2, advertising the connected networks, and disabling automatic summarization to ensure accurate route exchange. EIGRP was configured using Autonomous System number 100 and appropriate network statements with wildcard masks. The 'network' command was used to advertise connected networks, while 'no auto-summary' prevented classful route summarization. The 'show ip route' command was used to verify that routes were learned dynamically, and the 'ping' command was used to test end-to-end connectivity between devices. The successful ping results confirmed that both routing protocols allowed communication between all networks.

Conclusion

This assignment successfully demonstrated the configuration and management of routers using Cisco Packet Tracer. Basic router configuration tasks such as changing the hostname, configuring console access, and enabling login authentication were completed successfully. Static routing was then implemented to establish communication between different networks, and connectivity was verified using routing tables and ping tests.

Furthermore, dynamic routing protocols including RIP and EIGRP were configured and tested. The routing tables confirmed that routes were learned and exchanged correctly between routers, while successful ping results verified end-to-end communication across all networks. Through this assignment, a practical understanding of router configuration, IP addressing, static routing, and dynamic routing protocols was gained. Overall, the objectives of the assignment were achieved successfully, and all configured networks were able to communicate with each other as expected.